

Lecture 6: Social Preferences

Fairness, Reciprocity, and Why We Are Not Purely Selfish

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Opening Puzzle — Would You Reject This?

The Ultimatum Game — Your Decision

You and a stranger have been given **\$30** to split. The stranger decides how to split it. You either:

- **Accept:** You get the offered amount; the stranger keeps the rest.
- **Reject:** You both get **\$0**.

The stranger offers you \$5. Do you accept?

(Write your answer. Standard economics says: YES — $\$5 > \0 .)

Typical Result

In hundreds of experiments worldwide, **50% of subjects reject offers below \$3 out of \$10**. They willingly sacrifice their own payoff to punish unfair behaviour. Standard economics cannot explain this. Today we study why.

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2. **Ultimatum Game** — costly punishment of unfairness
3. **Dictator Game** — pure giving, no strategic motive
4. **Trust Game** — trust, trustworthiness, and economic growth
5. **Fehr-Schmidt Model** — inequality aversion (envy and guilt)
6. **Reciprocity and altruism** — giving and punishing
7. **Public goods games** — free-riding and altruistic punishment
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The Standard Assumption and the Evidence Against

Standard Economics

Agents maximise their *own* material payoff. Other people's payoffs enter only if they affect your own (rival goods, externalities).

Experiment	Nash Prediction	Typical Result
Ultimatum Game	Offer \$0.01; Responder accepts	Mean offer $\approx 40\%$; 50% reject $< 20\%$
Dictator Game	Give \$0	Mean giving $\approx 28\%$
Trust Game	Send \$0	Average sent $\approx 50\%$
Public Goods	Contribute \$0	Contribute 40–60% initially

These Are Not Errors — They Are Preferences

The gap between Nash predictions and actual behaviour is *systematic*, *stable*, and *replicable*. People have genuine preferences over others' outcomes. This is **social preferences**.

Rules

1. Pair up with your neighbour. Decide who is **Proposer** (P) and who is **Responder** (R).
2. P divides a hypothetical **10 tokens** between P and R. Write the offer to R (0–10).
3. R does not see P's offer yet. R writes a **minimum acceptable offer** (0–10).
4. If P's offer $\geq$ R's minimum: both get the amounts. If P's offer $<$ R's minimum: both get 0.

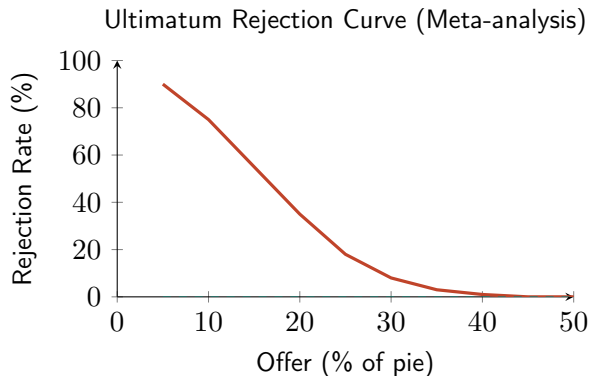
Data Collection

Show of hands: What did Proposers offer? What was the lowest accepted offer? Were any offers rejected? *Prof will compile results on the board.*

What to Expect

Modal offer: 4–5 tokens (40–50%). Rejections: common for offers below 2–3 tokens. Very few offer 0 or 1. Prediction: your class results will mirror the worldwide experimental literature.

Ultimatum Game — Worldwide Data



Meta-analysis (Oosterbeek et al., 2004): 37 studies. Mean offer: **37–45%**. Rejection for $<20\%$: **40–60%**. Near-zero: almost always rejected.

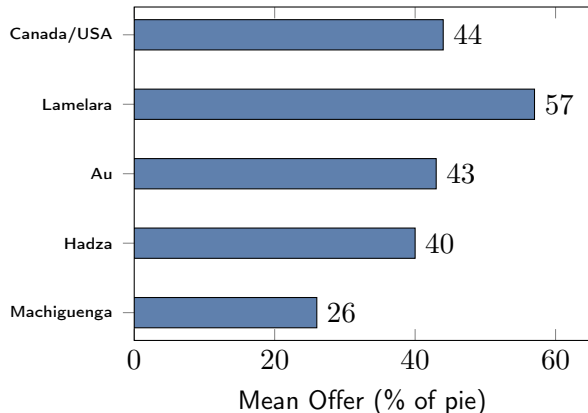
High Stakes Change Nothing

Slonim & Roth (Slovakia); Cameron (Indonesia): stakes up to 3 months' salary. Same pattern.

“In Search of Homo Economicus” — 15 Small-Scale Societies

Researchers ran ultimatum games in 15 traditional societies across 5 continents.

Mean Ultimatum Offer by Society



Interpretation

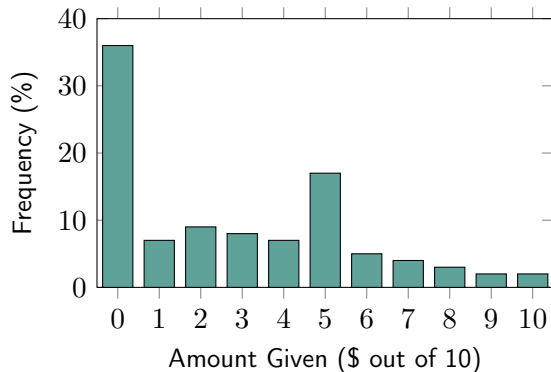
- **Machiguenga (Peru):** low offers, almost no rejections — social norms accept inequality.
- **Lamelara (Indonesia):** *hyper-fair* offers (57%); some OVER-generous offers rejected as embarrassing.
- **Canada/USA:** mean 44%, moderate rejection of low offers.

No society comes close to Nash equilibrium.
Pure self-interest is **never** the norm anywhere.

The Dictator Game — Pure Social Preferences

Responder **must accept** whatever Dictator offers. Any giving = *pure social preference* (altruism, guilt, inequality aversion).

Dictator Game Giving Distribution (Engel 2011)



Engel (2011) Meta-Analysis (616 papers):

- Mean giving: **28.3%**. **36%** give \$0. **17%** give 50%.
- Giving falls with anonymity and earned income.
- Giving rises with charity recipient and social observation.

The Trust Game (Investment Game)

Berg, Dickhaut & McCabe (1995)

1. Player A receives \$10. Can send any amount $X \in [0, 10]$ to B.
2. Amount **triples** in transit: B receives $3X$.
3. B can return any amount $Y \in [0, 3X]$ to A.

Nash prediction: B returns nothing $\Rightarrow$ A sends nothing (backward induction).

Result: A sends $\approx 50\%$; B returns $\approx 37\%$ of $3X$.

Why Does Trust Exist?

Sending is partly altruism (I care about B) and partly trust (I expect B to return). Cox (2004) separates these using an experimental design: altruism explains $\approx 50\%$, expected reciprocity $\approx 50\%$.

Trust Predicts Growth

Knack & Keefer (1997): 1 standard deviation increase in social trust $\Rightarrow +0.5$ pp annual growth. Canada trust level: **53%** (World Values Survey). Norway: **73%**. Brazil: **10%**. This gap predicts large differences in economic performance.

The Fehr-Schmidt Model (1999)

Utility Function

$$U_i(\mathbf{x}) = x_i - \alpha_i \max(x_j - x_i, 0) - \beta_i \max(x_i - x_j, 0)$$

- x_i : own payoff
- $\alpha_i \geq 0$: disutility from **disadvantageous inequality** (envy: “I have less than you”)
- $\beta_i \in [0, 1)$: disutility from **advantageous inequality** (guilt: “I have more than you”)
- Constraint: $\beta_i < 1$ (you never give away enough to fall below the other person)
- $\alpha_i \geq \beta_i$ (envy $\geq$ guilt)

Calibrated Parameters (Fehr & Schmidt, 1999)

Typical estimates: $\alpha \approx 0.5$ – 1.0 , $\beta \approx 0.2$ – 0.4 .

Being *behind* by \$10: costs $0.5 \times 10 = 5$ utils extra.

Being *ahead* by \$10: costs $0.25 \times 10 = 2.5$ utils extra.

Responder's Decision

Responder (type α_i) accepts offer s (fraction of pie = 1) if:

$$s \geq \alpha_i(1 - 2s)$$

Solving: $s \geq \frac{\alpha_i}{1 + 2\alpha_i}$

α_i	Min accept s	Min accept \$
0	0	\$0
0.5	0.25	\$2.50
1.0	0.33	\$3.33
4.0	0.44	\$4.44

Proposer's Decision

High- β proposer (guilt-averse) wants to offer close to 50% to minimise guilt. Low- β proposer offers just above the expected minimum acceptance threshold.

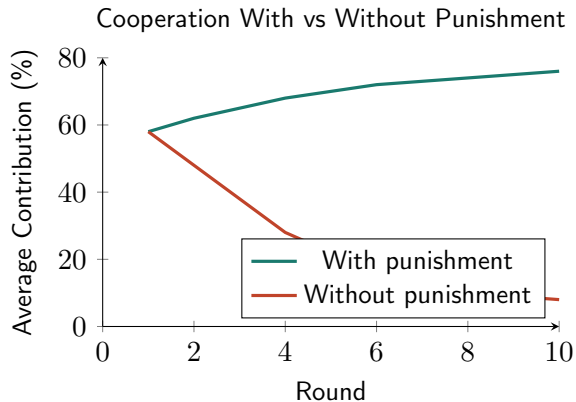
Result: diverse α - β population generates the observed distribution of offers and rejections.

Critique

Model assumes preferences over outcomes, not intentions. Rabin (1993): people respond to *perceived intentions*, not just payoffs. Fehr-Schmidt misses reciprocity entirely.

Altruistic Punishment — The Key Experiment

Fehr & Gächter (2002), *Nature*: Public goods game with and without a punishment option.



Key Finding

- **Without punishment:** cooperation $\rightarrow$ 0% by round 10.
- **With punishment:** cooperation rises, stays $>70\%$.
- Punishers pay even in one-shot games: **altruistic punishment**.

6,000+ citations — most cited experimental paper in economics.

[GAME] Public Goods Game

Rules

Each student secretly writes an amount **0–5 tokens** to contribute to the class pot. Prof multiplies total contributions by **1.5** and divides equally among all students.

If all contribute 5: each gets $1.5 \times 5 = 7.5$ tokens (gain!).

If you contribute 0, others contribute 5: you get free-rider share; others subsidise you.

Play 5 Rounds — Then Add Punishment

After round 5: each student can pay 1 token to *reduce a named free-rider's payoff by 3 tokens*. Watch what happens to contributions in rounds 6–10.

Prediction

Without punishment: contributions decline. With punishment: contributions rise. Free-riders are punished; norms enforced. This mirrors the real economy — norms work through costly enforcement.

Akerlof (1982) — Gift Exchange in Labour Markets

Firms that pay **above-market wages** receive **above-minimum effort** from workers — not because of threat of firing (rational), but because of *positive reciprocity*: the wage is perceived as a gift, and the worker reciprocates with effort.

Fehr, Kirchsteiger & Riedl (1993) — Lab Evidence

Experimental labour market: firms post wage offers, workers choose effort. Anonymous, one-shot. Higher wages $\Rightarrow$ higher effort — *even with no reputation effects*. Pure reciprocity at work.

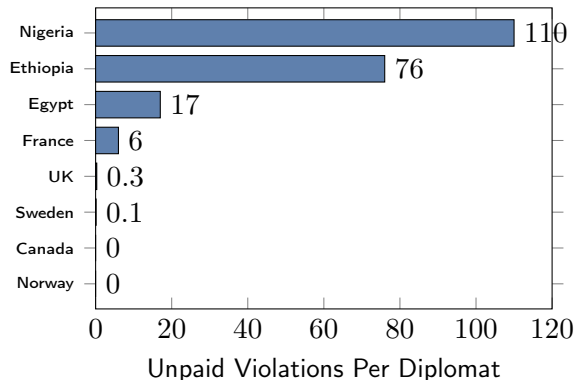
Macroeconomic Consequence: Wage Rigidity

Workers resist nominal wage *cuts* more than equivalent real wage cuts via inflation, because cuts violate the implicit fairness norm of the wage relationship. This generates downward nominal wage rigidity and explains involuntary unemployment.

Fisman & Miguel (2007) — UN Parking Violations

Before 2002, UN diplomats in New York had **complete diplomatic immunity** from parking fines. A natural experiment: any violation of parking norms was *purely voluntary*.

UN Parking Violations (Fisman & Miguel 2007)



What This Shows

Home-country corruption norms **persist** even when there is no legal penalty. Diplomats from low-corruption countries (Norway, Canada): near-zero violations. High-corruption countries: hundreds of violations per diplomat. *Social norms are internalized — not just imposed by law.*

Discussion Questions

- Q1. “You rejected the \$5 offer in the opening puzzle. Does this make you irrational (you sacrificed \$5) or rational (you enforced a fairness norm that benefits society)?”
- Q2. “If workers have inequality aversion ($\beta > 0$), does a CEO-to-worker pay ratio of 200:1 create efficiency losses through resentment and lower effort? How would you test this?”
- Q3. “Canada scores 53% on the social trust question. What policies could raise this? Does higher trust always benefit economic outcomes?”

Key Takeaways — Lecture 6

- ✓ **Pure self-interest fails:** people give away money, reject unfair offers, and punish free-riders at personal cost.
- ✓ **Ultimatum:** mean offer $\approx 40\%$; rejections common below 20%. Same in high stakes.
- ✓ **Dictator:** mean giving $\approx 28\%$ even when anonymous and no retaliation possible.
- ✓ **Trust game:** trust and trustworthiness exist; social trust predicts GDP.
- ✓ **Fehr-Schmidt:** utility decreases from both disadvantageous (α) and advantageous (β) inequality.
- ✓ **Altruistic punishment:** people pay to punish free-riders; enforces cooperation norms.
- ✓ **Applications:** efficiency wages, wage rigidity, corruption, social norms.

Next: Lecture 7: Behavioural Game Theory — Level-k thinking, the beauty contest, cooperation in repeated games, and why Nash equilibrium often fails in the real world.